Unit-6 Graph Theory

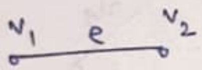
A graph written as $G(V, E)$ consists of two components

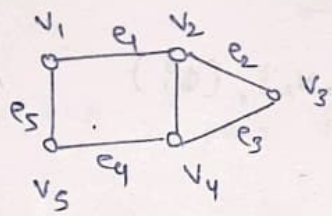
(i) The set of vertices V, also called points or nodes

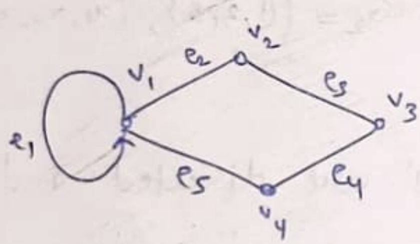
(ii) The set of edge E, also called lines are arcs, connecting pair of vertices

Each pair of vertices, connected by an edge, is called the end points or end vertices.

Example ⇒







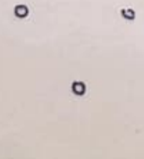
Types of Graphs ⇒

① Trivial Graph ⇒ Consisting only one vertex and no edge.

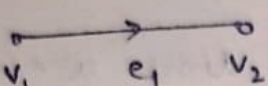
Ex ⇒ 

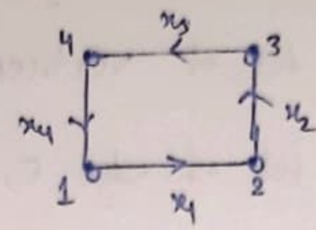
② Null Graph ⇒ Consisting n vertices and no edge.

Eg ⇒



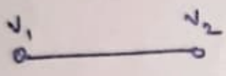
(3) Directed Graph ⇒ A graph consist the direction of edges then this is called directed graph.



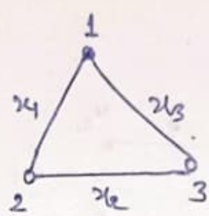


$G_1 = (\{v_1, v_2\}, \{e_1\})$ $\qquad$ $G_2 = (\{1,2,3,4\}, \{x_1, x_2, x_3, x_4\})$

(4) Undirected graph ⇒ A graph which is not directed.



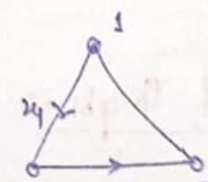
$G_1 = (\{v_1, v_2\}, \{\phi\})$



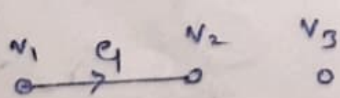
$G_2 = (\{1,2,3\}, \{x_1, x_2, x_3\})$

(5) mixed Graph ⇒ If some edges are directed and some are undirected.

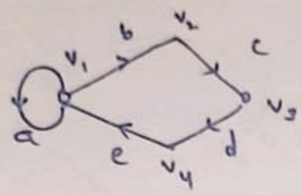
$G_1 = (\{1,2,3\}, \{x_1, x_2\})$



(6) Isolated ~~Graph~~ Vertex ⇒ A vertex v which is not connected with any vertex of a graph G, by an edge, is called an isolated vertex.



(7) **Self loop** ⇒ An edge having the same vertex as both its end vertices is called a self loop.



~~$V = \{v_1, v_2, v_3, v_4\}$~~

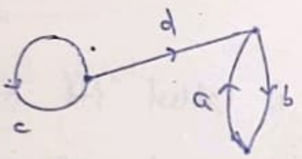
~~$E = f$~~

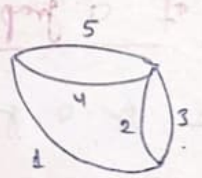
(8) **Proper Edge** ⇒ An edge which is not self loop is called proper edge.

Eg. In above diagram ⇒ b, c, d, e.

(9) **multiple edges / Parallel edges** ⇒ If a pair of vertices is joined by more than one edge, then these edges are called parallel edges.

Eg.

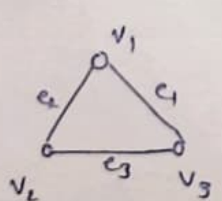


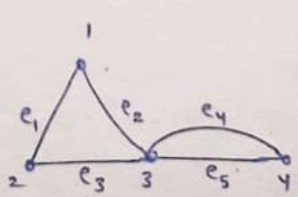


Parallel ⇒ (a, b)

(2, 3) & (4, 5)

(10) **Simple Graph** ⇒ A graph having neither self loop nor parallel edge is called a simple graph otherwise called multi-graph.



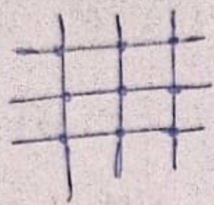


(Simple Graph)

(multi-Graph)

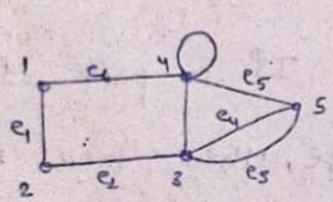
(11) **Finite & Infinite Graph** ⇒ In a graph $G = (V, E)$, if the set V and E both finite it is called finite graph otherwise an infinite graph.

Eg



(12) **Pseudo Graph** ⇒ A graph contain both self loop & multi-edge is called pseudo graph

Eg ⇒

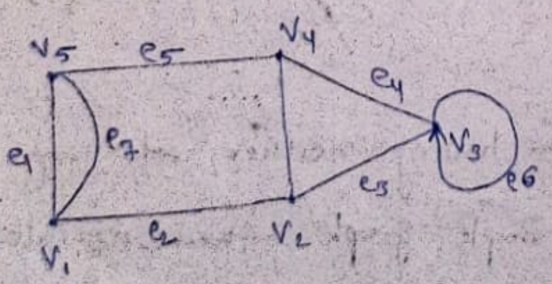


(13) **Order and ~~degree~~ size of a graph** ⇒

order ⇒ The no. of vertices in a graph is called its order.

~~degree~~ size ⇒ The no. of edges in a graph is called its size

Eg ⇒

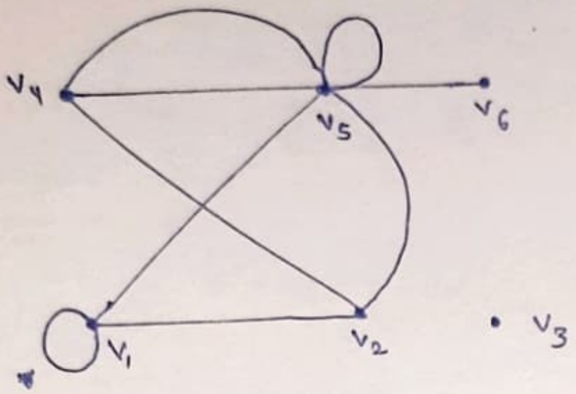


order = 5

~~degree~~

size = 7.

(14) Degree of a vertex ⇒ The no. of edges incident on a vertex v of a graph is called degree of the vertex.



$d(v_1) = 4$
$d(v_2) = 3$
$d(v_3) = 0$
$d(v_4) = 3$
$d(v_5) = 7$
$d(v_6) = 1$

(15) Pendant vertex ⇒ A vertex of degree one is ~~call~~ known as pendant vertex.

Eg ⇒ In above diagram, v_6 is a pendant vertex.

(16) Isolated vertex ⇒ A vertex of degree "0" is known as isolated vertex.

Eg. ⇒ In above diagram, v_3 is an isolated vertex.

Types of a graph →

① **Complete Graph** ⇒ A simple connected graph is said to be complete if each vertex is connected to the every other vertex.

* The complete graph of n- vertices is denoted by K_n.

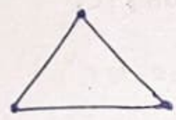
* The size of $K_n = {}^{n}C_2 = \frac{n(n-1)}{2}$

Eg ⇒ K_1 ⇒

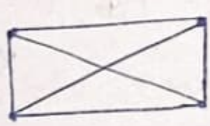
K_5 ⇒

K_2 ⇒

K_3 ⇒



K_4 ⇒

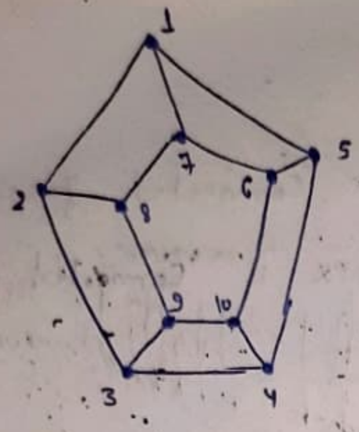


② **Regular Graph** ⇒ A simple graph is called regular if every vertex of the graph has the same degree.

NOTE ⇒ A regular graph is called n- regular graph if every vertex is of degree n.

Eg ⇒ Construct a 3- regular graph of 10- vertices.

Q.



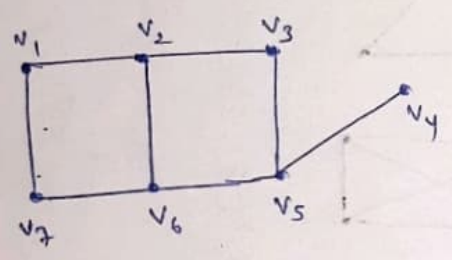
No. of vertices = 10

degree of each vertices = 3

⇒ 3- regular graph.

③ **Bigraph / Bipartite Graph** ⇒ If the vertex set V of graph G can be divided into non-empty disjoint subsets X and Y in such a way that edge of G has one end in X and one end in Y, then G is called bipartite.

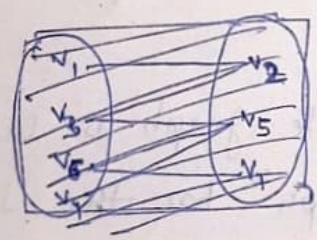
Eg → ①



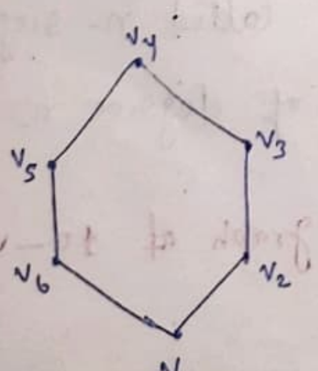
$V = \{v_1, v_2, \dots v_7\}$

$X = \{v_1, v_6, v_3, v_4\}$

$Y = \{v_7, v_2, v_5\}$



②



$V = \{v_1, v_2, \dots v_6\}$

$X = \{v_1, v_3, v_5\}$

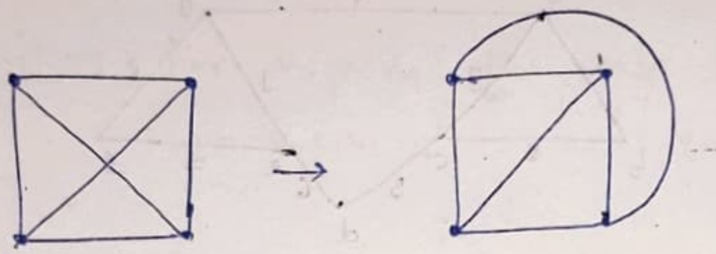
$Y = \{v_2, v_4, v_6\}$

~~(4) Complete bipartite Graph ⇒~~

Planer Graph ⇒ (KTU-2024)

A graph is called planer if it can be drawn in the plane without intersecting any edge.

Eg.

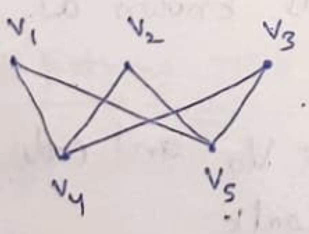


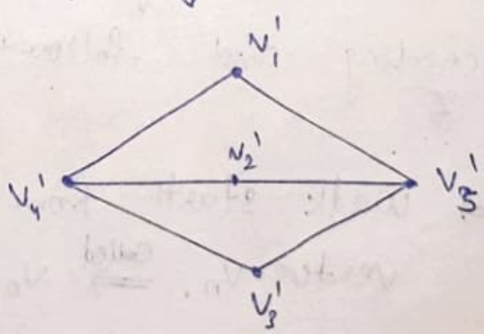
It is a planer graph b/c it can be drawn w/o crossings.

Isomorphic Graph ⇒ (KTU-2024)

Two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are said to be isomorphic to each other if there exists a bijection mapping f from V_1 to V_2.

Eg. ⇒ $G(V, E)$ $G'(V', E')$



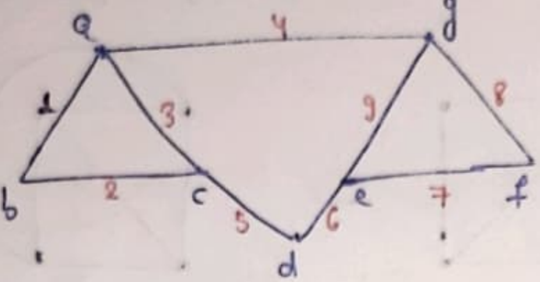


G & G' are isomorphic b/c $f: V \to V'$ is bijection.

Weighted Graph ⇒

A graph $G = (V, E)$ is called a weighted graph, when each edge is assigned a +ve real no.

eg.



edge$(a,b) = 1$, edge $(b,c) = 2$, edge$(c,a) = 3$

edge $(c,d) = 5$, .. etc

Walk ⇒ ~~let G = VE~~

let $G = (V, E)$ be a graph. A finite alternating sequence of vertices and edges denoted by W, beginning and ending with vertices.

$$W = V_0\, e_1\, V_1\, e_2\, V_2\, e_3\, V_3\, e_4 \text{------} V_{n-1}\, e_n\, V_n \quad (n \geq 0).$$

Such that each edge is incident with the vertices preceeding and following it is known as the walk.

NOTE ⇒ walk starts from vertex V_0 and ends with vertex V_n, $\xRightarrow{\text{called}}$ $V_0 - V_n$ walk

$V_0 \rightarrow$ origin vertex

$V_n \rightarrow$ Terminal vertex

Length of a walk ⇒

The no. of edges (not all different necessary) in a walk is called its length.

NOTE ⇒ walk of length zero is called trivial walk.

Open walk ⇒ for $v_0 \neq v_n$ the walk

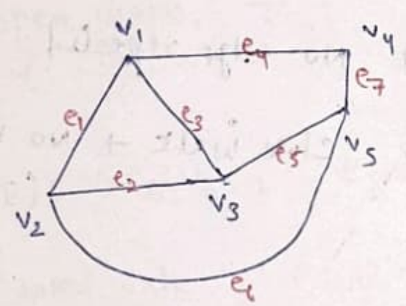
$w = v_0 e_1 v_1 e_2 \ldots v_{n-1} e_n v_n$ ⇒ open walk.

Closed walk ⇒ $v_0 = v_n$ ⇒ Closed walk.

Trail ⇒ An open walk in which no edge is repeated.

Path → An open walk in which no vertex appears more than once is called a path.

Eg ⇒



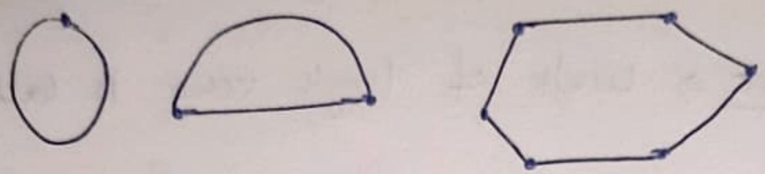
$w = v_1 e_3 v_3 e_2 v_2 e_6 v_5 e_7 v_4$

~~NOTE ⇒ Path~~

Length of path ⇒ The no. of edges in a path is called the ~~edge~~ length of the path.

Circuit ⇒ A closed walk in which no edge appears more than once, is called a circuit.

eg



Cycle ⇒ A closed path is called cycle.

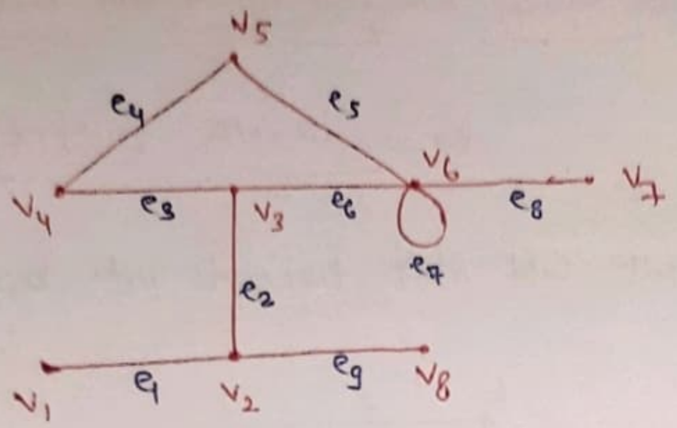
NOTE ⇒ In a closed path the end vertices are same.

NOTE :

- Trail ⇒ Open walk + No edge repeated
- Path ⇒ open walk + No vertex repeated
- Circuit ⇒ closed walk + No edge repeated
- cycle ⇒ close path ⇒ close walk + No vertex repeated (If last same only)

Eulerian Path ⇒

Q:



(1) $w_1 = v_1 e_1 v_2 e_9 v_8 e_9 v_2 e_2 v_3 e_6 v_6 e_6 v_3 e_3 v_4$

Ans ⇒ open walk, No trail, No path
(L=7)

L → length

(2) $w_2 = v_5 e_5 v_6 e_7 v_6 e_8 v_7$

Ans → open walk, Trail, No Path
(L=3)

(3) $w_3 = v_1 e_1 v_2 e_2 v_3 e_6 v_6 e_8 v_7$

Ans ⇒ open walk, Trail, Path.
(L=4)

(4) $w_4 = v_4 e_3 v_3 e_6 v_6 e_7 v_6 e_5 v_5 e_4 v_4$

Ans ⇒ Closed walk, Closed Trail, No Closed Path, Circuit
(L=5)

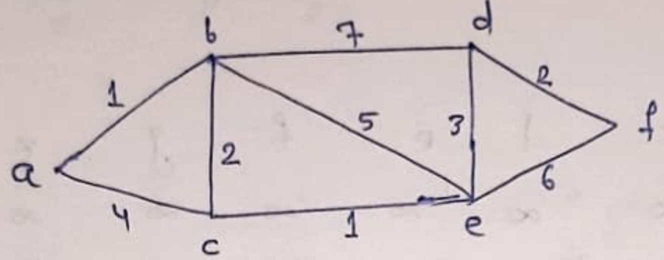
(5) $w_5 = v_3 e_3 v_4 e_4 v_5 e_5 v_6 e_6 v_3$

Ans ⇒ Closed walk, Closed Trail, Closed Path, cycle.
(L=4)

Shortest Path in weighted graph ⇒

Dijkstra's Algorithm ⇒

Q: Find the shortest Path b/w the vertices a & f.



Sol:

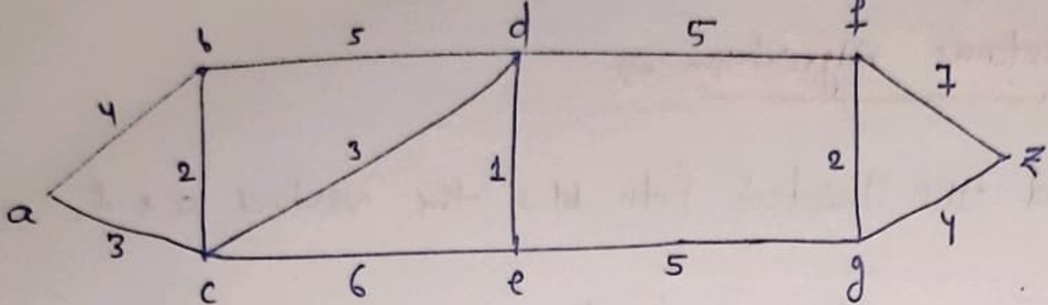
a	b	c	d	e	f
$\boxed{0}$ ✓	∞	∞	∞	∞	∞
$\boxed{0}$	$\boxed{1}$ ✓	4	∞	∞	∞
$\boxed{0}$	$\boxed{1}$	$\boxed{3}$ ✓	8	6	∞
$\boxed{0}$	$\boxed{1}$	$\boxed{3}$	8	$\boxed{4}$ ✓	∞
$\boxed{0}$	$\boxed{1}$	$\boxed{3}$	$\boxed{7}$ ✓	$\boxed{4}$	10
$\boxed{0}$	$\boxed{1}$	$\boxed{3}$	$\boxed{7}$	$\boxed{4}$	$\boxed{9}$ ✓

Shortest Path ⇒ a – b – c – e – d – f

~~sho~~ length of shortest Path = 9

Q: find shortest path b/w a and z. (RTU 2024)

10-m.



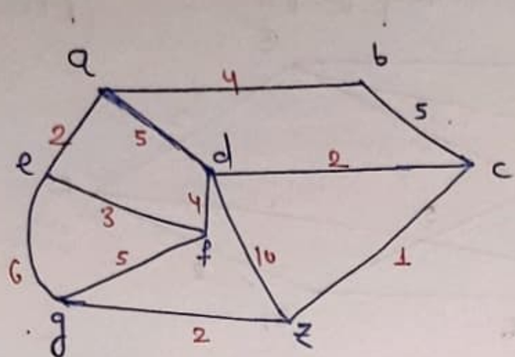
Sol:

a	b	c	d	e	f	g	z
$\boxed{0}$ ✓	∞	∞	∞	∞	∞	∞	∞
$\boxed{0}$	4	$\boxed{3}$ ✓	∞	∞	∞	∞	∞
$\boxed{0}$	$\boxed{4}$ ✓	$\boxed{3}$	6	9	∞	∞	∞
$\boxed{0}$	$\boxed{4}$	$\boxed{3}$	$\boxed{6}$ ✓	9	∞	∞	∞
$\boxed{0}$	$\boxed{4}$	$\boxed{3}$	$\boxed{6}$	$\boxed{7}$ ✓	11	∞	∞
$\boxed{0}$	$\boxed{4}$	$\boxed{3}$	$\boxed{6}$	$\boxed{7}$	$\boxed{11}$ ✓	12	∞
$\boxed{0}$	$\boxed{4}$	$\boxed{3}$	$\boxed{6}$	$\boxed{7}$	$\boxed{11}$	$\boxed{12}$ ✓	18
$\boxed{0}$	$\boxed{4}$	$\boxed{3}$	$\boxed{6}$	$\boxed{7}$	$\boxed{11}$	$\boxed{12}$	$\boxed{16}$ ✓

Shortest Path ⇒ a − c − d − e − g − z

Length ⇒ 16

Q: Shortest Path = ?



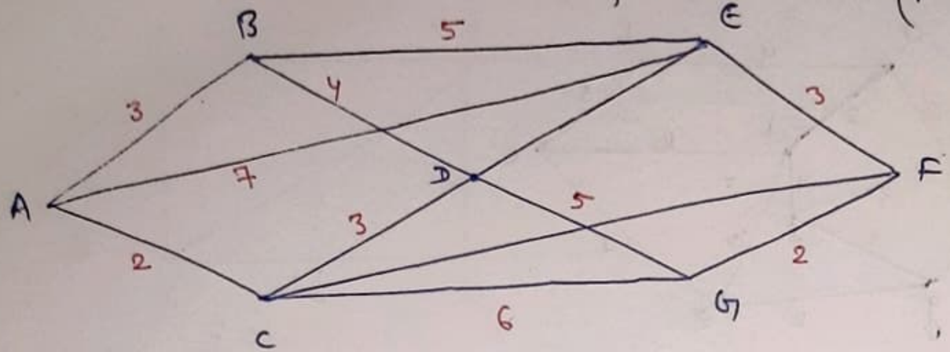
Sol:

a	b	c	d	e	f	g	z
$\boxed{0}$	∞	∞	∞	∞	∞	∞	∞
$\boxed{0}$	4	∞	5	$\boxed{2}$	∞	∞	∞
$\boxed{0}$	$\boxed{4}$	∞	5	$\boxed{2}$	5	8	∞
$\boxed{0}$	$\boxed{4}$	9	$\boxed{5}$	$\boxed{2}$	5	8	∞
$\boxed{0}$	$\boxed{4}$	7	$\boxed{5}$	$\boxed{2}$	$\boxed{5}$	8	15
$\boxed{0}$	$\boxed{4}$	$\boxed{7}$	$\boxed{5}$	$\boxed{2}$	$\boxed{5}$	8	15
$\boxed{0}$	$\boxed{4}$	$\boxed{7}$	$\boxed{5}$	$\boxed{2}$	$\boxed{5}$	8	$\boxed{8}$
$\boxed{0}$	$\boxed{4}$	$\boxed{7}$	$\boxed{5}$	$\boxed{2}$	$\boxed{5}$	$\boxed{8}$	$\boxed{8}$

Shortest Path ⇒ a – d – c – z

length ⇒ 8

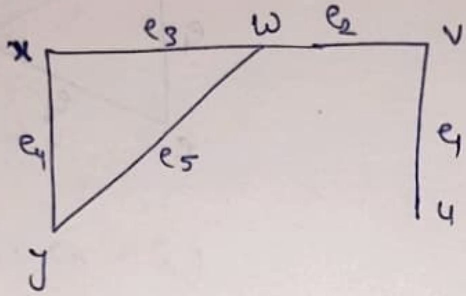
Q: Use Dijkstra's algo, find shortest dist. of all vertices from vertex A for the follwg graph. (KTU-2022)



~~Hamiltonian~~ Eulerian Paths & Circuits ⇒

Eulerian Path ⇒ A path in a graph is said to be an Eulerian path if it traverses each edge in the graph once and only once.

Eg ⇒



→ open

→ Each Edge → No Repeat will appear

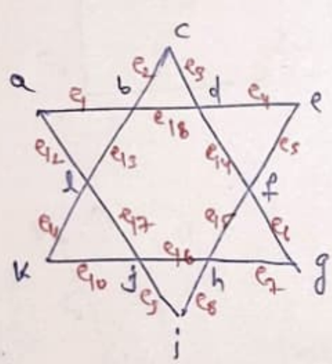
→ ~~no vertex repeat.~~

→ vertex may rep.

$W = u\,e_1\,v\,e_2\,w\,e_5\,y\,e_4\,x\,e_3\,w$

Eularian Circuit ⇒ A circuit in a graph is said to be an Eularian circuit if it ~~travers~~ traverses each edge in the graph once & only once

Eg. ⇒



→ Close

→ Each ~~edge~~ edge → come

→ No rep.

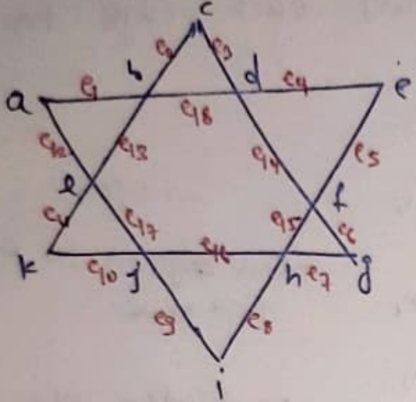
→ vertex → may be rep.

$W =$ ~~a-b-c-d~~

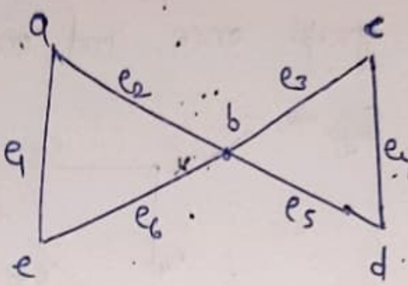
a - b - c - d - e - f - g - h - i - j - k - l - b - d - f - h - j - l

Eulerian Graph ⇒ A connected graph which contain an Eularian Circuit is called Eularian Graph.

Eg ⇒ g ①



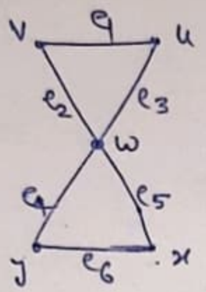
g. ②



Hamiltonian Path ⇒

A path which contain every vertex of a graph G exactly once is called H.P.

Eg

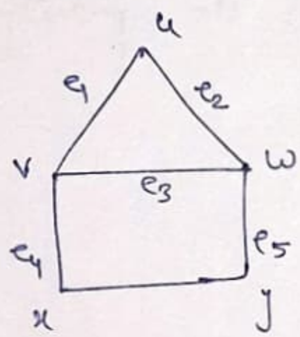


$W = u e_1 v e_2 w e_5 x e_6 y$

Hamiltonian Circuit ⇒

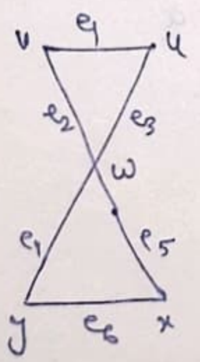
A circuit that passes through each of vertices in a group G exactly one except the starting vertex and end vertex is called H.C.

Eg ⇒



$w = u e_1 v e_4 x e_6 y e_5 w e_2 u$

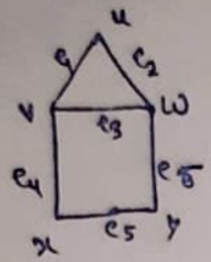
but



then $u e_1 v e_2 w e_5 x e_6 y e_4 w e_3 u$ is not Ham. Circuit b/w vertex w is repeated.

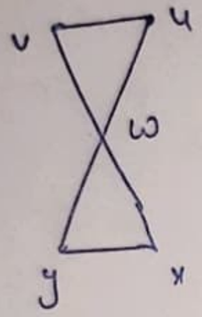
Hamiltonian Graph → A connected graph which contain Ham. Circuit is called Ham. Graph.

Ex



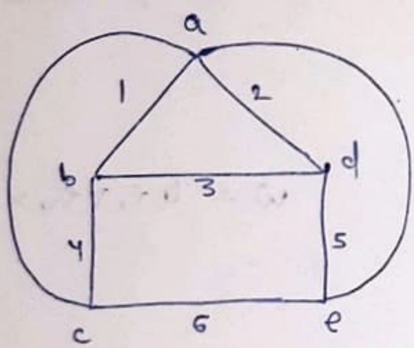
⇒ Ham. grap

But



⇒ is not Ham. graph.

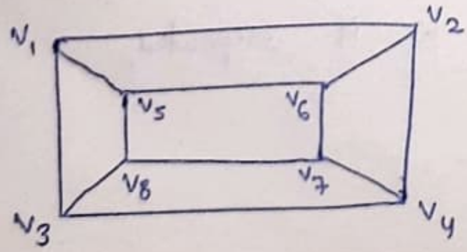
Q: Determine a min. Ham. Circuit.



Sol: Ham. Circuit ⇒ a 1 b 4 c 6 e 5 2 a.

(covers all vertices)

Q: Discuss the graph shown below.



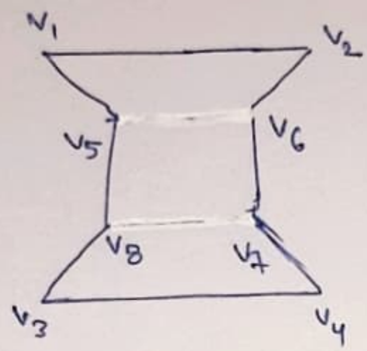
ⓐ Draw H.C. and H. Path

ⓑ Is it Ham. Graph ?

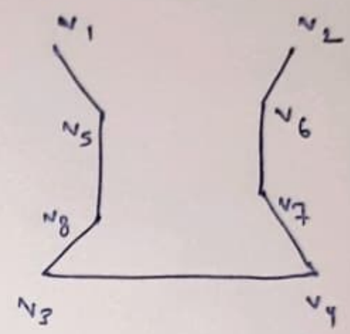
ⓒ Is it Euler Graph ?

Sol:

ⓐ H.C



H.P.



all vertices are covered (may close or not)

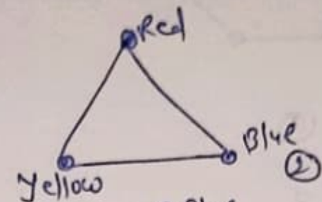
ⓑ yes, it covers all vertices so it H.G.

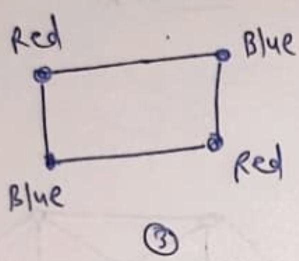
ⓒ No, by covering all edges, vertex will repeat

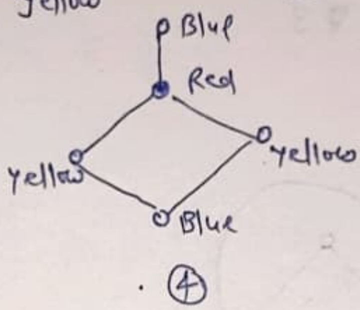
Graph Coloring ⇒

Painting all the vertices of a graph with color such that no two adjacent vertices have the same color is called coloring of a graph.

E.g.



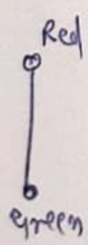




① ⇒ 2-Chromatic Graph
② ⇒ 3- " — " -
③ ⇒ 2 - " ——— " —
④ ⇒ 3 - —" ——— " ———

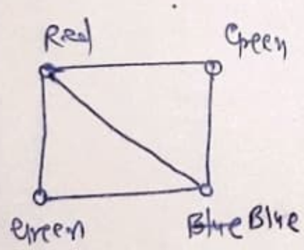
Chromatic No. ⇒ The least no. of colors required to coloring of a graph "G" is called chromatic no.

Eg ⇒



2-Chromatic no.

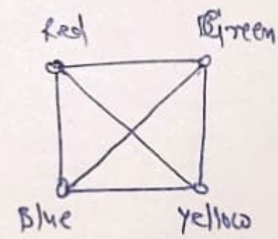
$\chi(G) = 2$



3 - C.N.

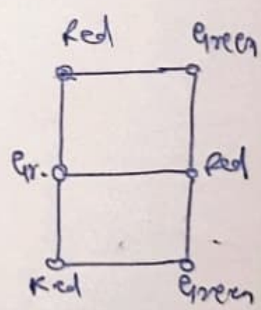
$\chi(G) = 3$

↳ (RTU-2024)



4 - C.N.

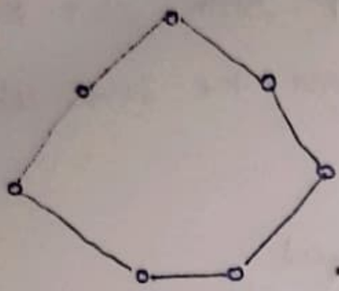
$\chi(G) = 4$



2 - Ch. No.

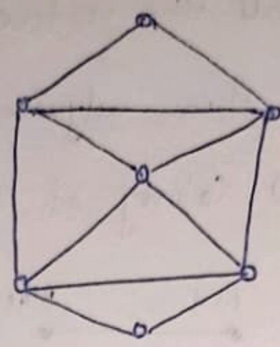
$\chi(G) = 2$

Q: Determine Ch. No of following graph



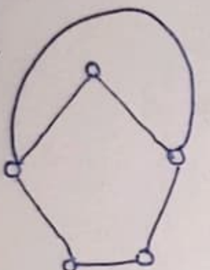
ⓐ

Ans ⇒ 3



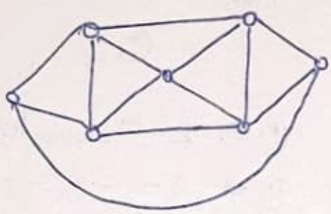
ⓑ

Ans ⇒ 3.



ⓒ

Ans ⇒ 3.

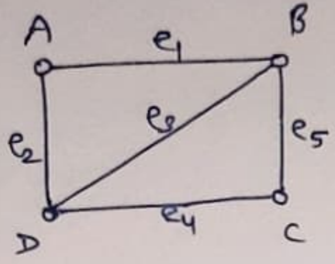


ⓓ

Ans ⇒ 4

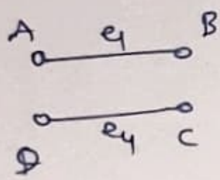
Matching graph → A matching graph is a subgraph of a graph G where there are no edges adjacent to each other.

Eg ⇒

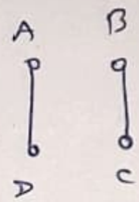


Matching graphs of the above graph is —

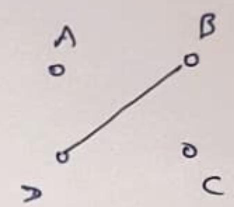
(1)



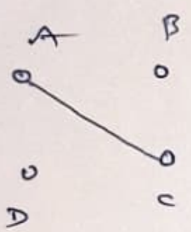
(2)



(3)



(4)



(5)